Equation Automata

How a traffic simulation discovered the Ising critical temperature?

by

GÖKTUĞ İSLAMOĞLU

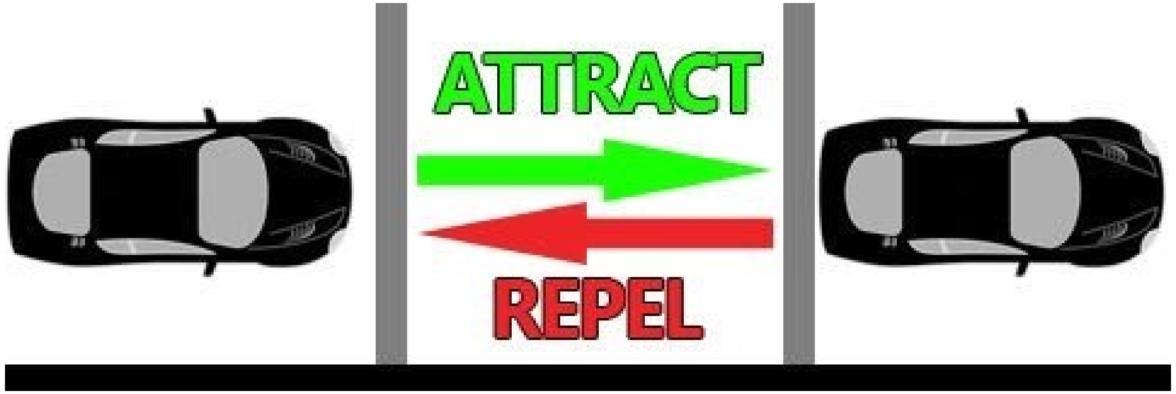
goktugislamoglu@gmail.com

MAGNETISM IN TRAFFIC

Attraction: drive forward to take the position of the car ahead

Repulsion: keeping distance so a crash doesn't happen

Attraction-repulsion forces



Model: Equation Automaton

Introduced to Cellular Automata: NECSI Courses on Complexity

Question:

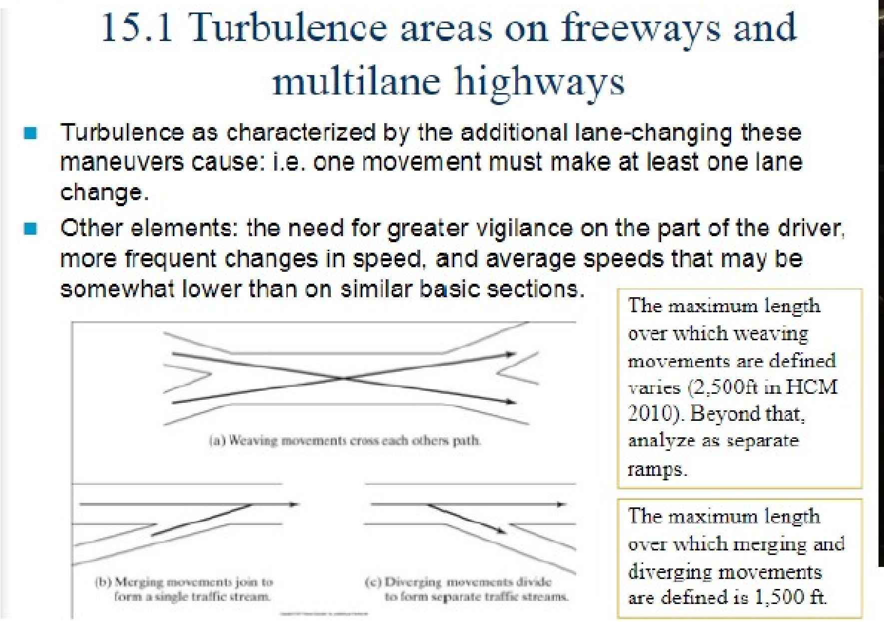
Accidents cause traffic buildup. Can traffic buildup cause accidents?

1. https://github.com/goektug/Equation-Automata/

BAD ROAD DESIGN: CRASHES

D100 roadway in the now decommissioned Ataturk Airport area:

└ Squeezed by BRT road on one side
└ Squeezed by side road on the other side



15.1 Turbulence areas on freeways and multilane highways

- Turbulence as characterized by the additional lane-changing these maneuvers cause: i.e. one movement must make at least one lane change.
- Other elements: the need for greater vigilance on the part of the driver, more frequent changes in speed, and average speeds that may be somewhat lower than on similar basic sections.

The maximum length over which weaving movements are defined varies (2,500ft in HCM 2010). Beyond that, analyze as separate ramps.

The maximum length over which merging and diverging movements are defined is 1,500 ft.

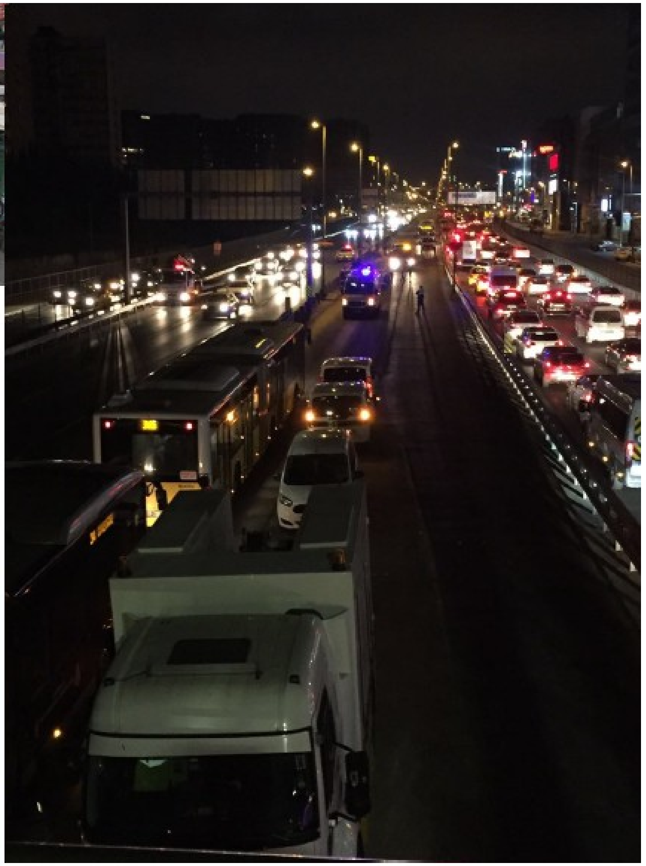
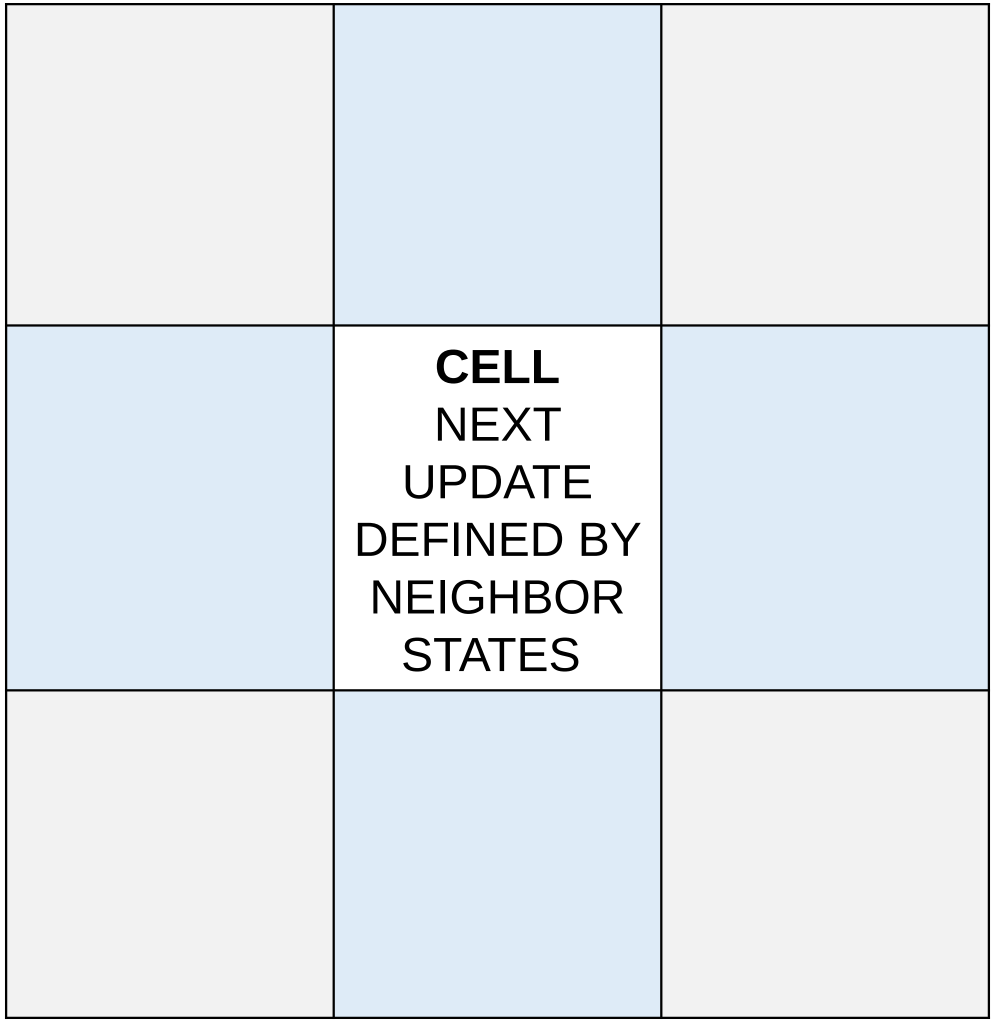


Photo 1 - Google Maps Street View – Şirinevler D100 Roadway and Side Road, Joining Ahead Without Access Road
Photo 2 – Goktug Islamoglu, Taken en Route to Airport, Şirinevler Metrobüs Stop. BRT Operations Stopped Due to Car Entering Protected Road
Image 1 - Traffic Engineering, Roger P. Roess / Elena S. Prassas / William R. McShane, Prentice Hall, Pearson .

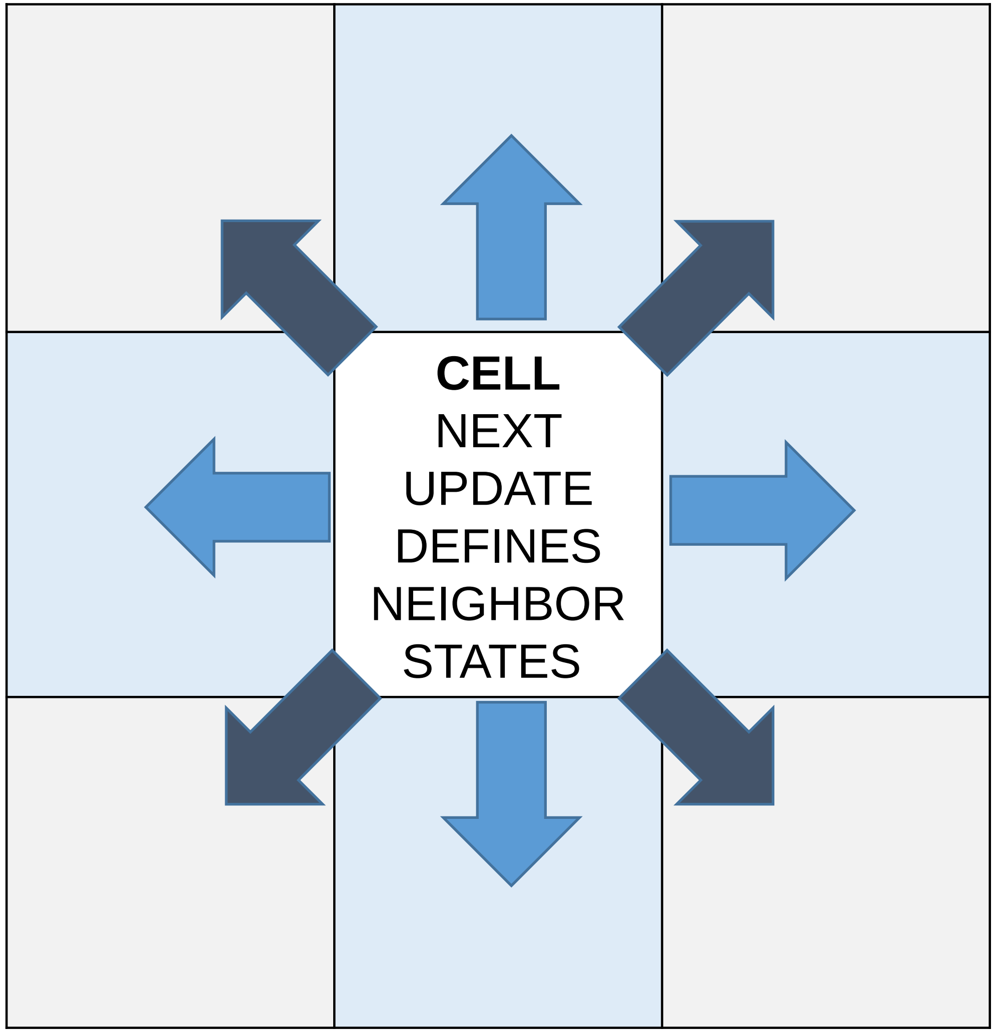
Conventional Cellular Automaton Update



A cellular automaton is a search function around a cell.

Update rule is the determination of a cell’s state based on the states of its neighboring cells following a rule.

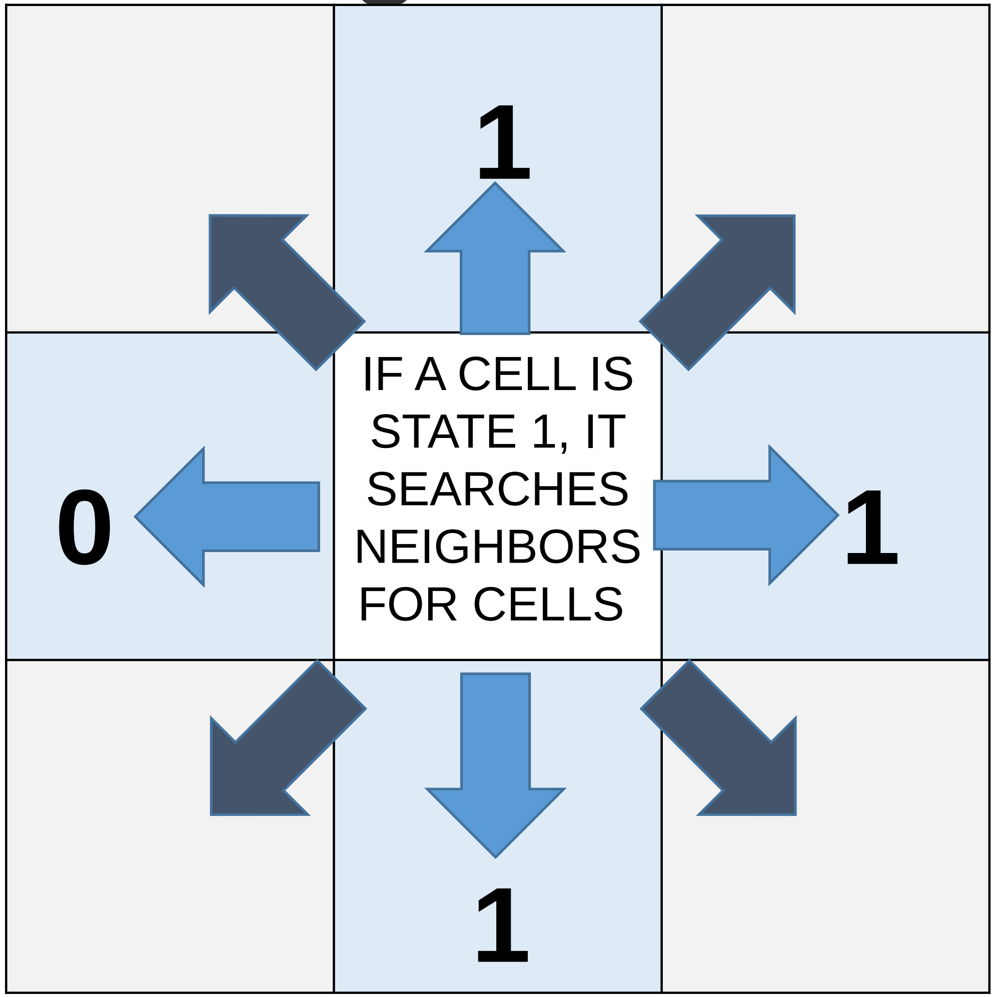
1st Rule: Inverse Cell Update



To achieve driving motion, update of a cell is reversed. Cell's state dictates its neighborhood.

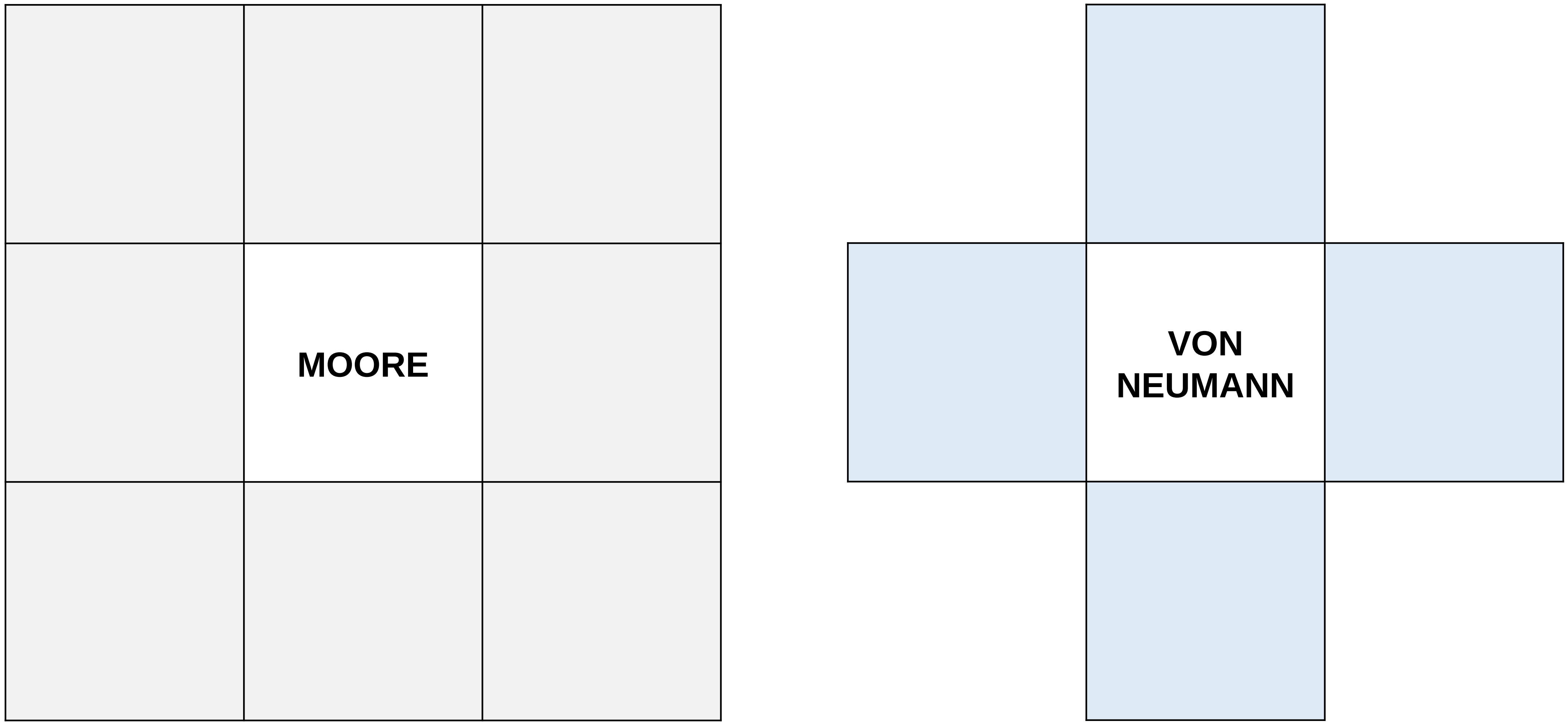
```
elif c[x, y] == 1:
    array1.append(c[x, y])
    for z in range(-1, 2):
        # block generation from randomly distributed points
```

Driving motion for neighboring cells

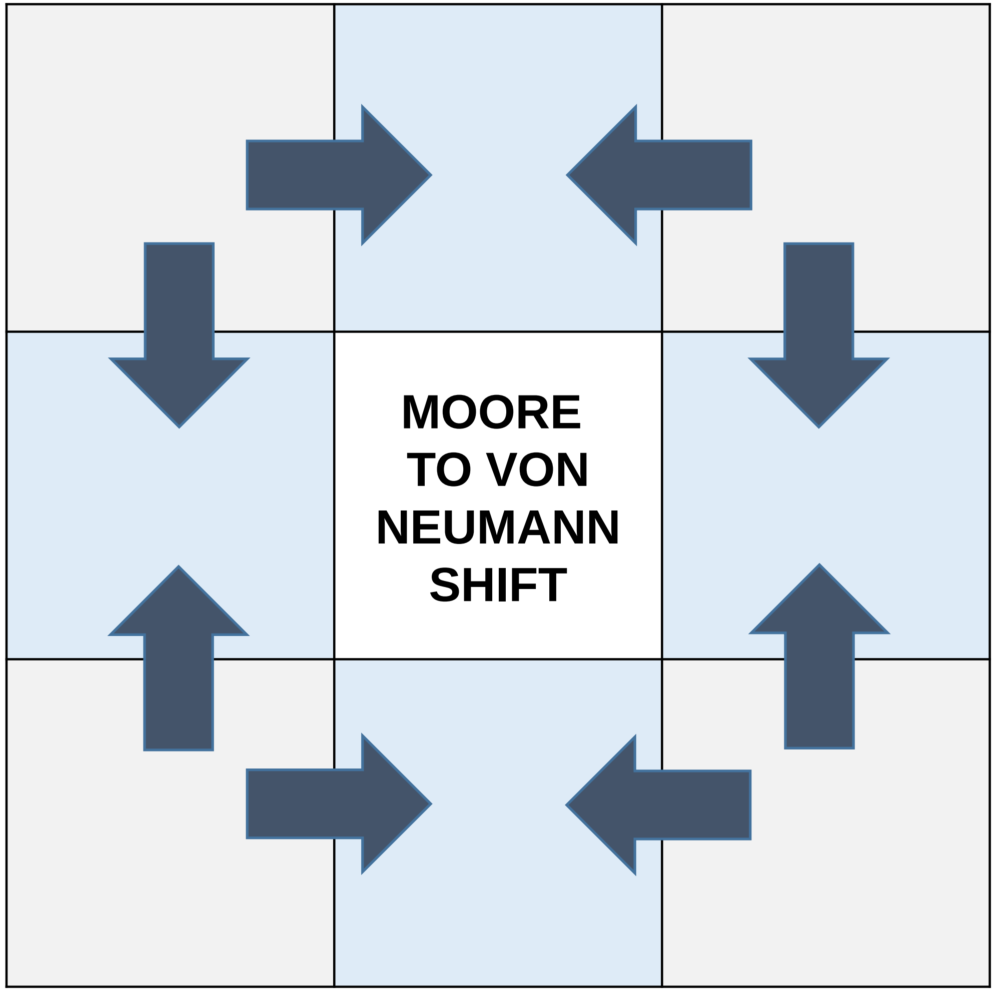



```
elif c[x, y] == 1:
    array1.append(c[x, y])
    for z in range(-1, 2):
        # block generation from randomly distributed points
        m = number_of_upper_neighbors(x, y)
        if m == 1:
            nc[x, (y + 1) % L] = 1
        n = number_of_lower_neighbors(x, y)
        if n == 1:
            nc[x, (y - 1) % L] = 1

        k = number_of_right_neighbors(x, y)
        if k == 0 and (m <= 1 or n <= 1):
            nc[(x + 1) % L, (y + z) % L] = 1

        l = number_of_left_neighbors(x, y)
        if l == 1 and (m > 1 or n > 1):
            nc[(x - 1) % L, (y + z) % L] = 0
```

Moore vs Von Neumann Neighborhoods



2nd Rule: Moore to Von Neumann Shift



To achieve criticality, competition between Moore cells and von Neumann cells are needed.

```
 g = number_of_Moore_neighbors(x, y) #CA tuning
 if c[x, y] == 0:
     nc[x, y] = 0 if g <= 6 else 1
     array0.append(c[x, y])

h = number_of_Neumann_neighbors(x, y) #CA tuning
if h >= 1:
    nc[x, y] = 1 if g <= 6 else 0
```

2nd Rule: Neighborhood Configuration

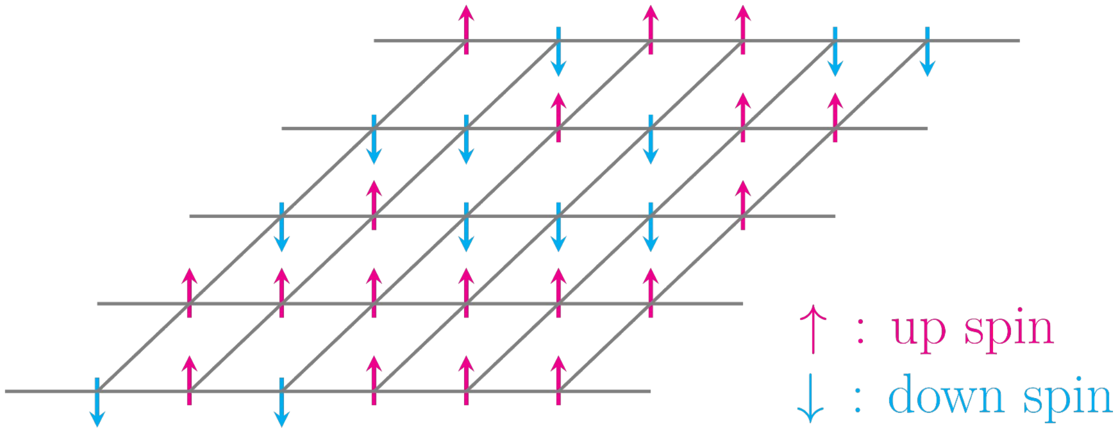
1	**1**	**1**
0	**TUNING THE NEIGHBOR NUMBERS**	**0**
1	**1**	**1**

Based on cell’s value, Moore and von Neumann neighborhoods are tuned. Critical values are 6 for both.

```
g = number_of_Moore_neighbors(x, y) #CA tuning
if c[x, y] == 0:
    nc[x, y] = 0 if g <= 6 else 1
    array0.append(c[x, y])

h = number_of_Neumann_neighbors(x, y) #CA tuning
if h >= 1:
    nc[x, y] = 1 if g <= 6 else 0
```

Ising Model

A simple model of magnetism: when interacting spins in neighboring sites "agree", they have a lower energy, and vice versa. System tends to low energy, but temperature may change.



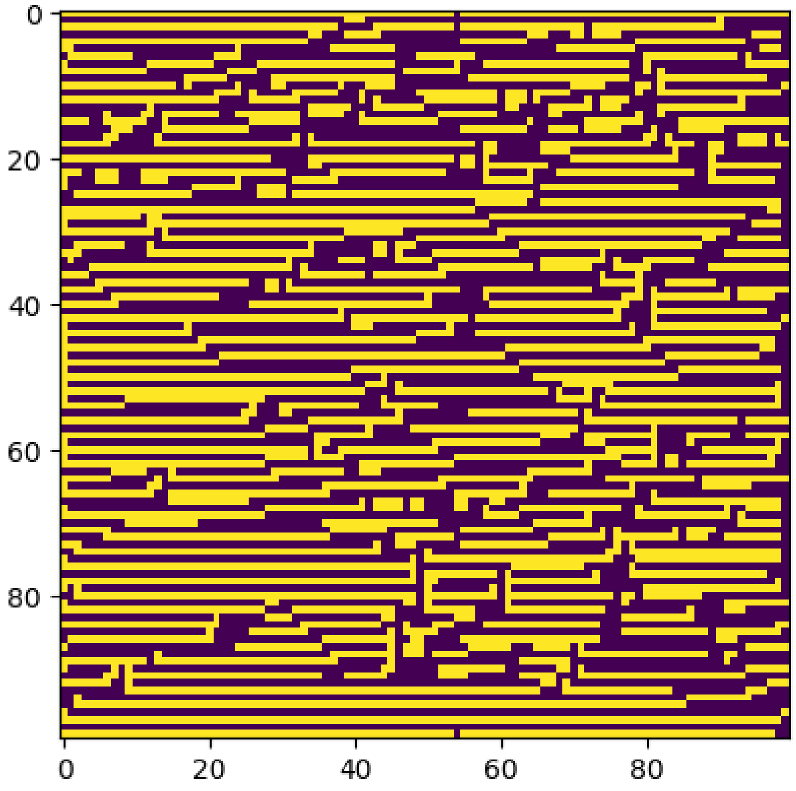
For dimensions greater than 1, there exists a phase transition.

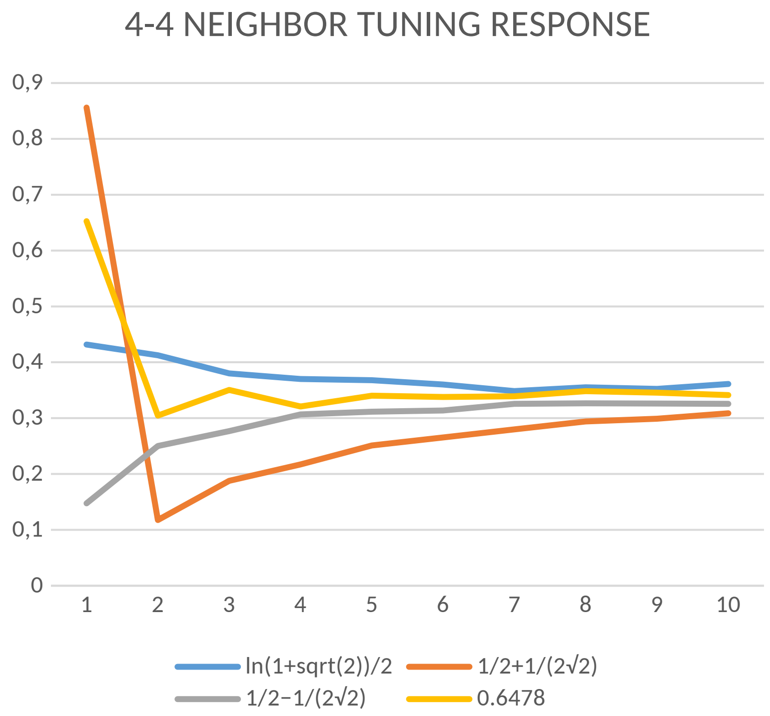
1. https://en.wikipedia.org/wiki/Ising_model

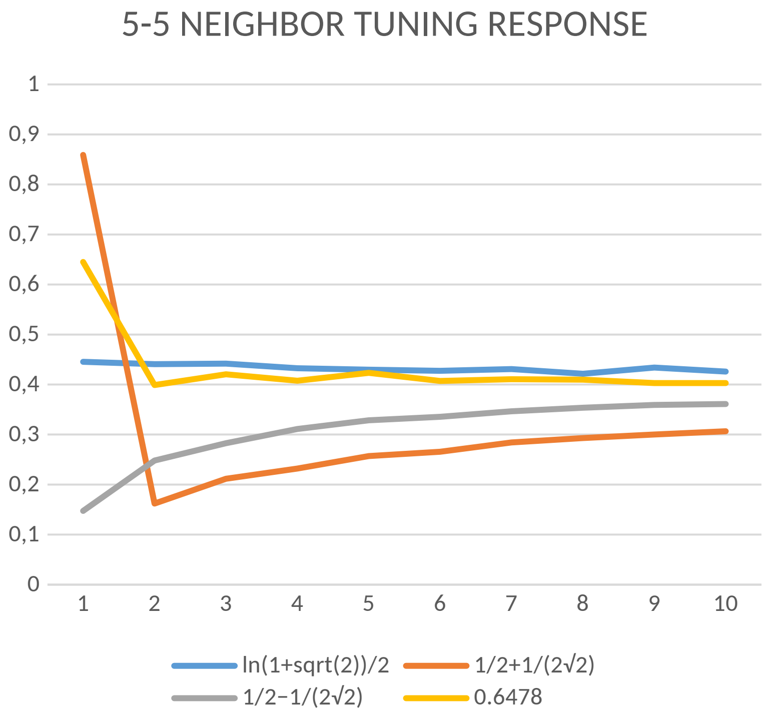
Idea

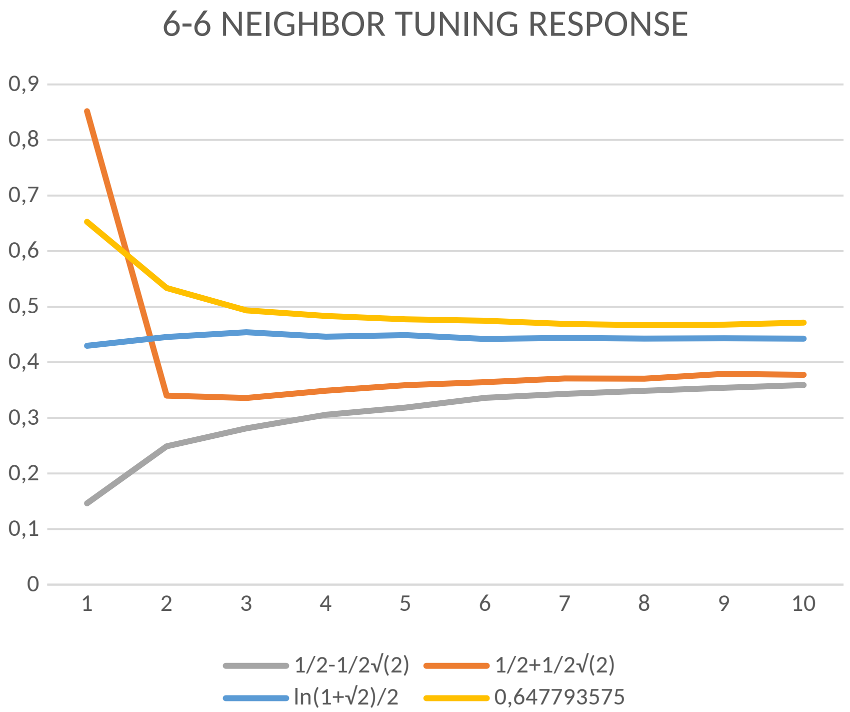
Inducing ferromagnetism by extending the driving motion to criticality

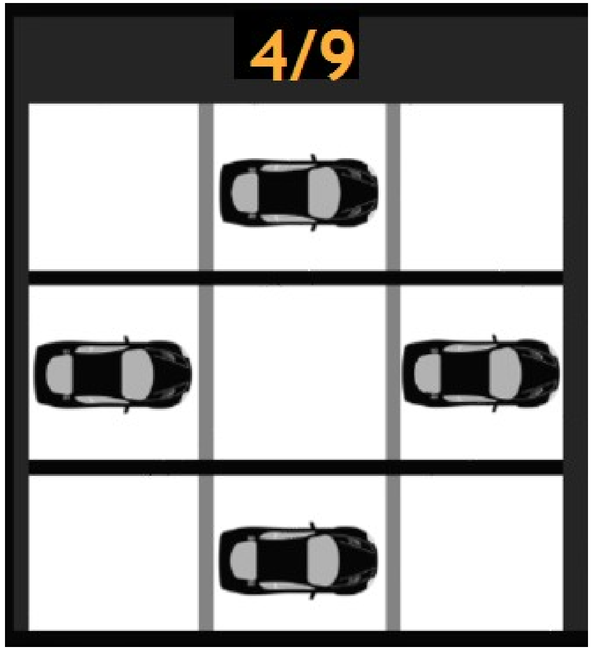
Hypothesis: Can a 2D Ising Model be projected onto 1D linear chain?

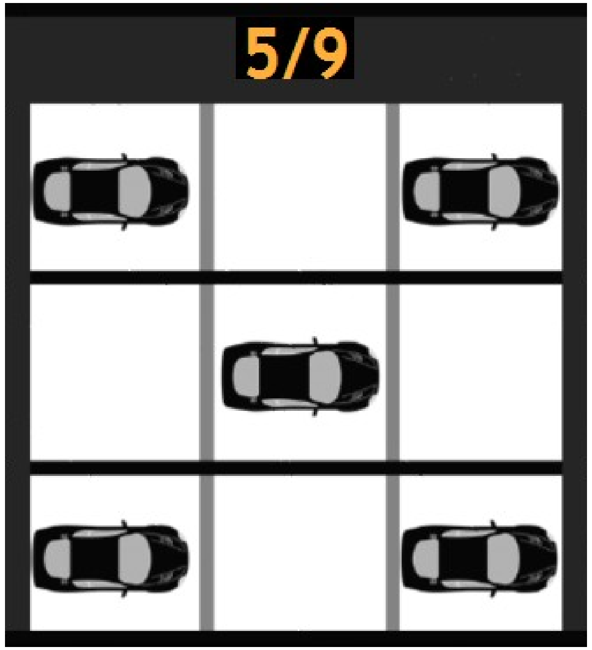



4-4 NEIGHBOR TUNING RESPONSE
0,9
0,8
0,7
0,6
0,5
0,4
0,3
0,2
0,1
0
1 2 3 4 5 6 7 8 9 10
ln(1+sqrt(2))/2
1/2+1/(2√2)
1/2−1/(2√2)
0.6478


5-5 NEIGHBOR TUNING RESPONSE
1
0,9
0,8
0,7
0,6
0,5
0,4
0,3
0,2
0,1
0
1 2 3 4 5 6 7 8 9 10
ln(1+sqrt(2))/2
1/2+1/(2√2)
1/2−1/(2√2)
0.6478


6-6 NEIGHBOR TUNING RESPONSE
0,9
0,8
0,7
0,6
0,5
0,4
0,3
0,2
0,1
0
1 2 3 4 5 6 7 8 9 10
1/2-1/2√(2)
1/2+1/2√(2)
ln(1+√2)/2
0,647793575


4/9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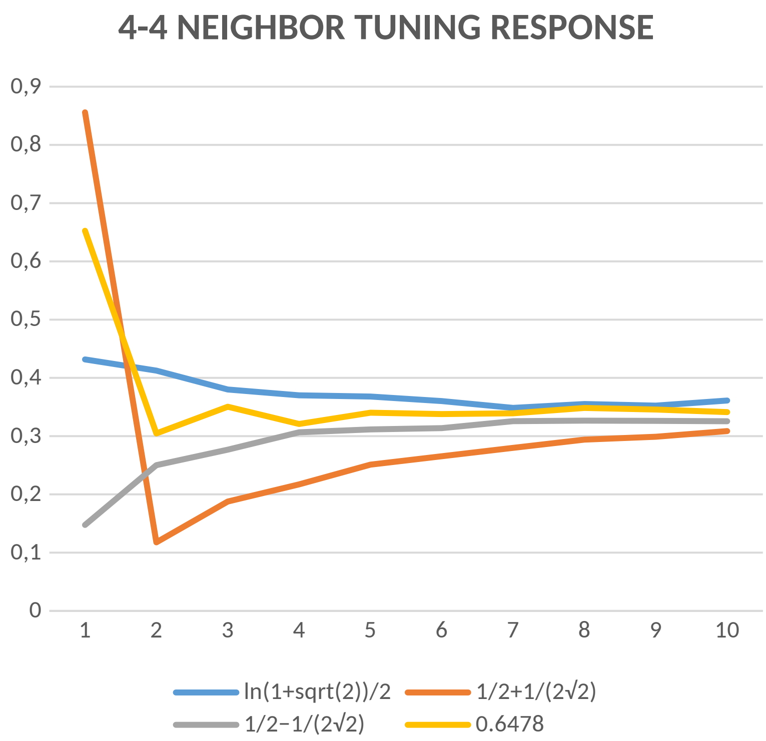
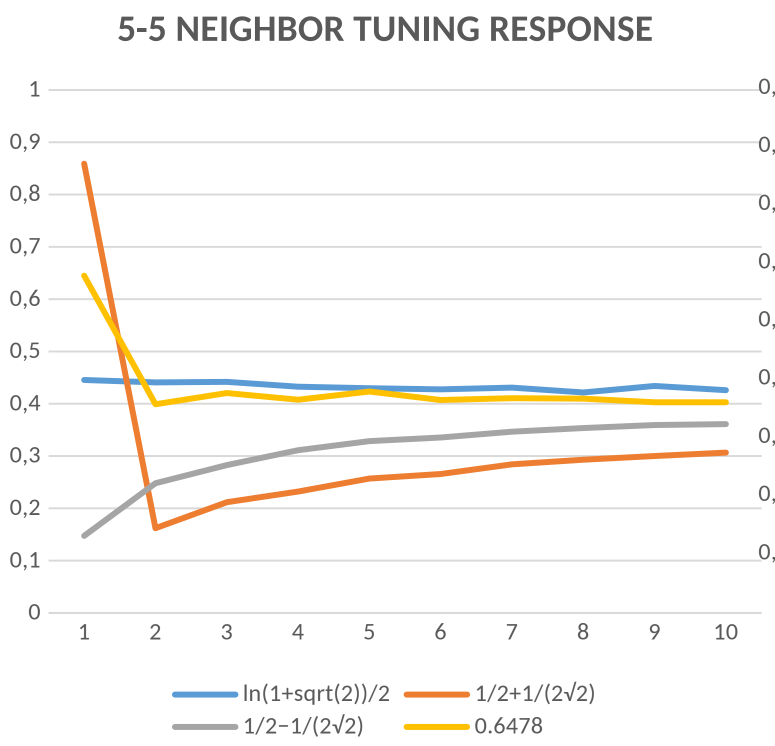
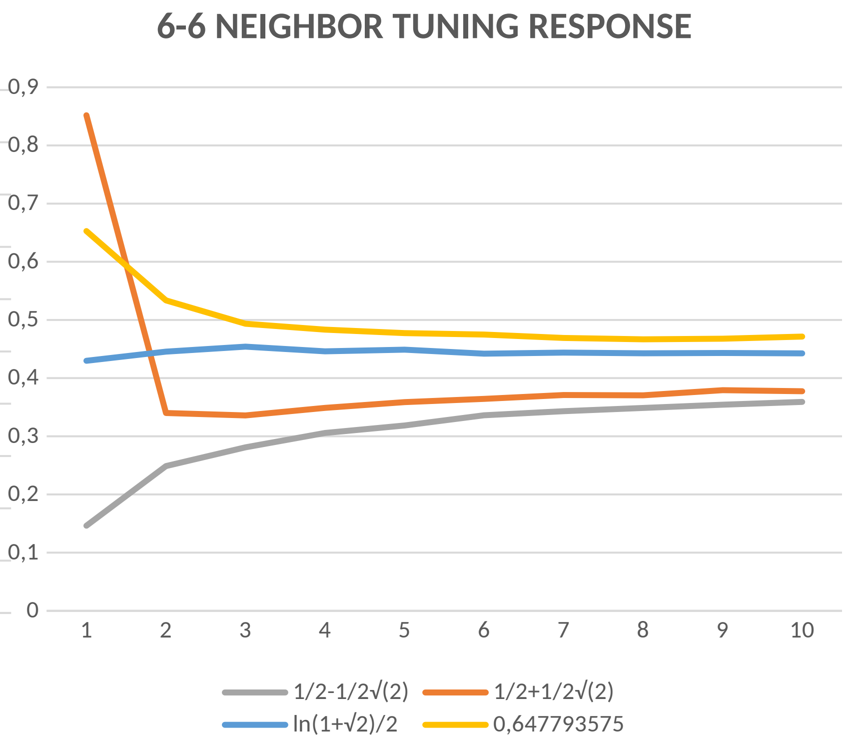
6/9

Neighbor Tuning for Ising Criticality



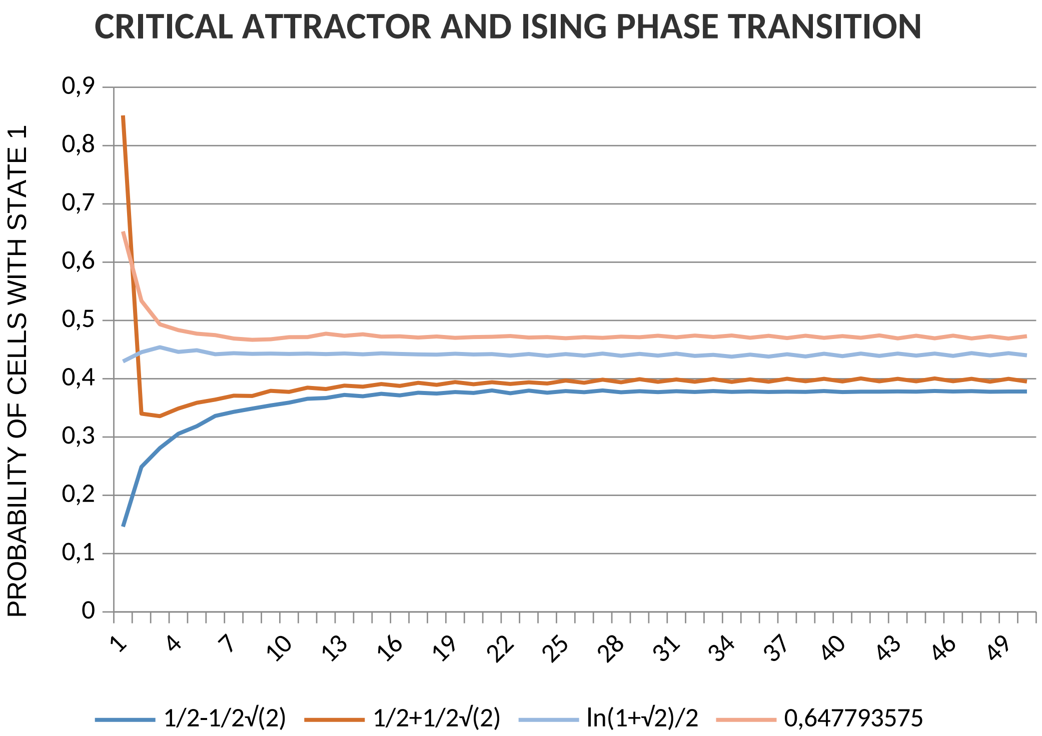


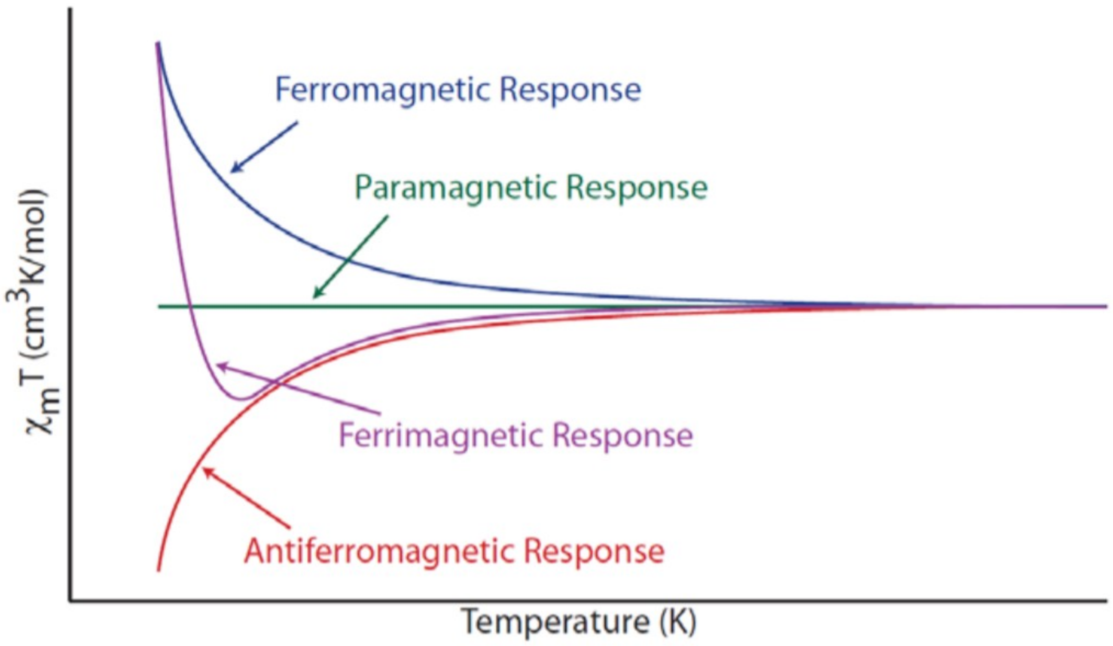


Tuning for **4** and **5** neighbors has **Inverse Ising critical temperature** as the highest state 1 cell count.

For **6** neighbors however, there is another maximum susceptibility, corresponding to ferromagnetism.

Ising Model Susceptibility





2D Ising square lattice model’s **critical inverse temperature** is the **paramagnetic response**.

Upper criticality is weak ferromagnetism (**ferrimagnetism**).

Lower criticality is **antiferromagnetic** behavior.

Right Figure: http://bh.knu.ac.kr/~leehi/index.files/MPMS_HIL.pdf

3rd Rule: Critical Cellular Automata

$$g/8 > (1 - p) * p$$

$$p^2 - p + 1/8 = 0$$

$$p = \frac{1}{2} \pm 1/(2\sqrt{2})$$

Left: 2D criticality condition broken down to its roots by assigning **1** to Moore neighborhood count g and creating an equation.

```
if g / 8 > (1 - p) * p:  # coupling function
    nc[(x + 1) % L, y] = 1
elif g / 8 < (1 - p) * p:
    nc[(x - 1) % L, y] = 1
else:
    nc[x, y] = 1
```

Evolution of Criticality

CLOCKWISE FROM TOP:

1. Neighbor update without criticality

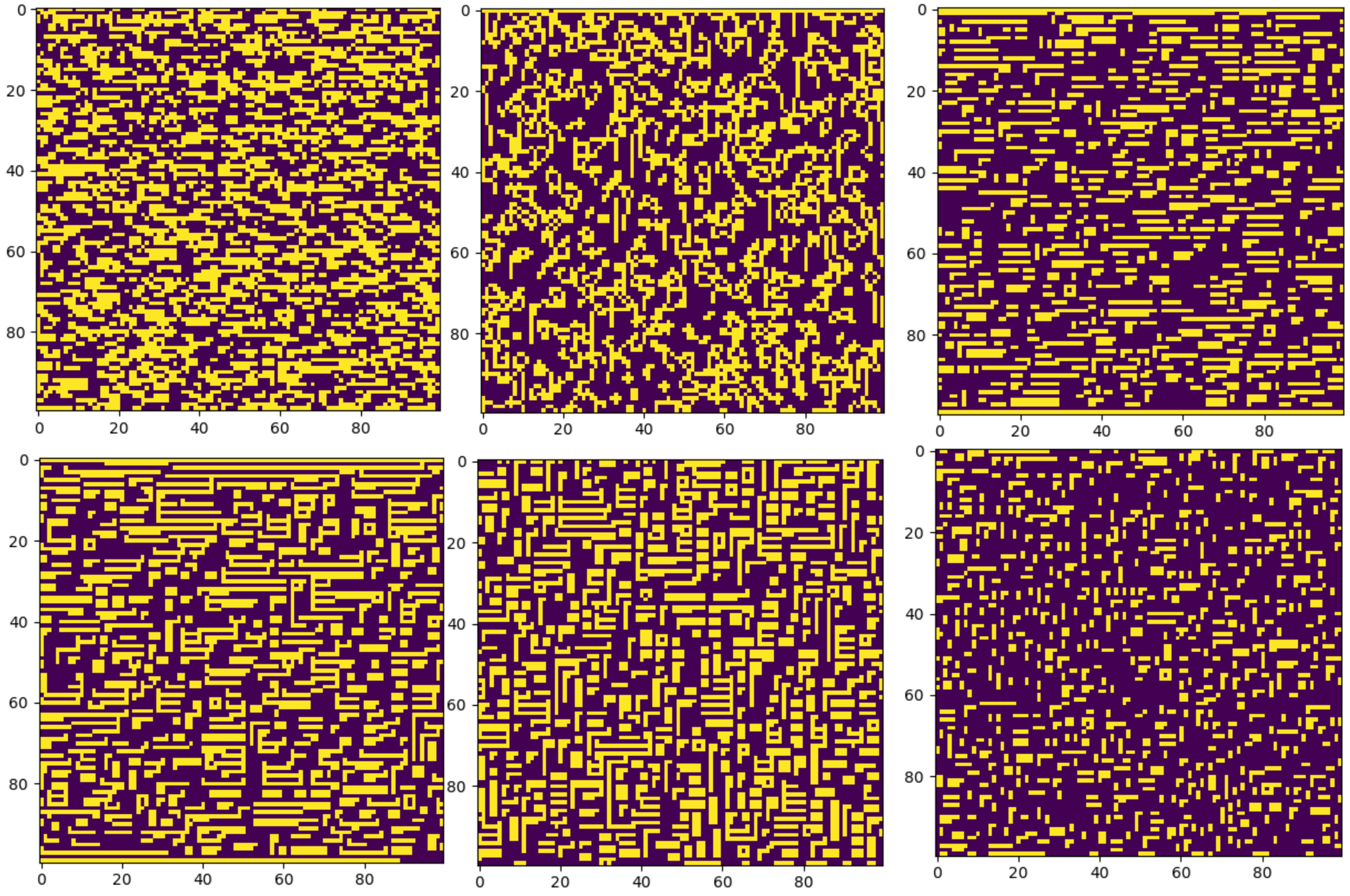
2. Criticality without neighbor update

3. Uncritical state above ½ ± 1/(2√2)

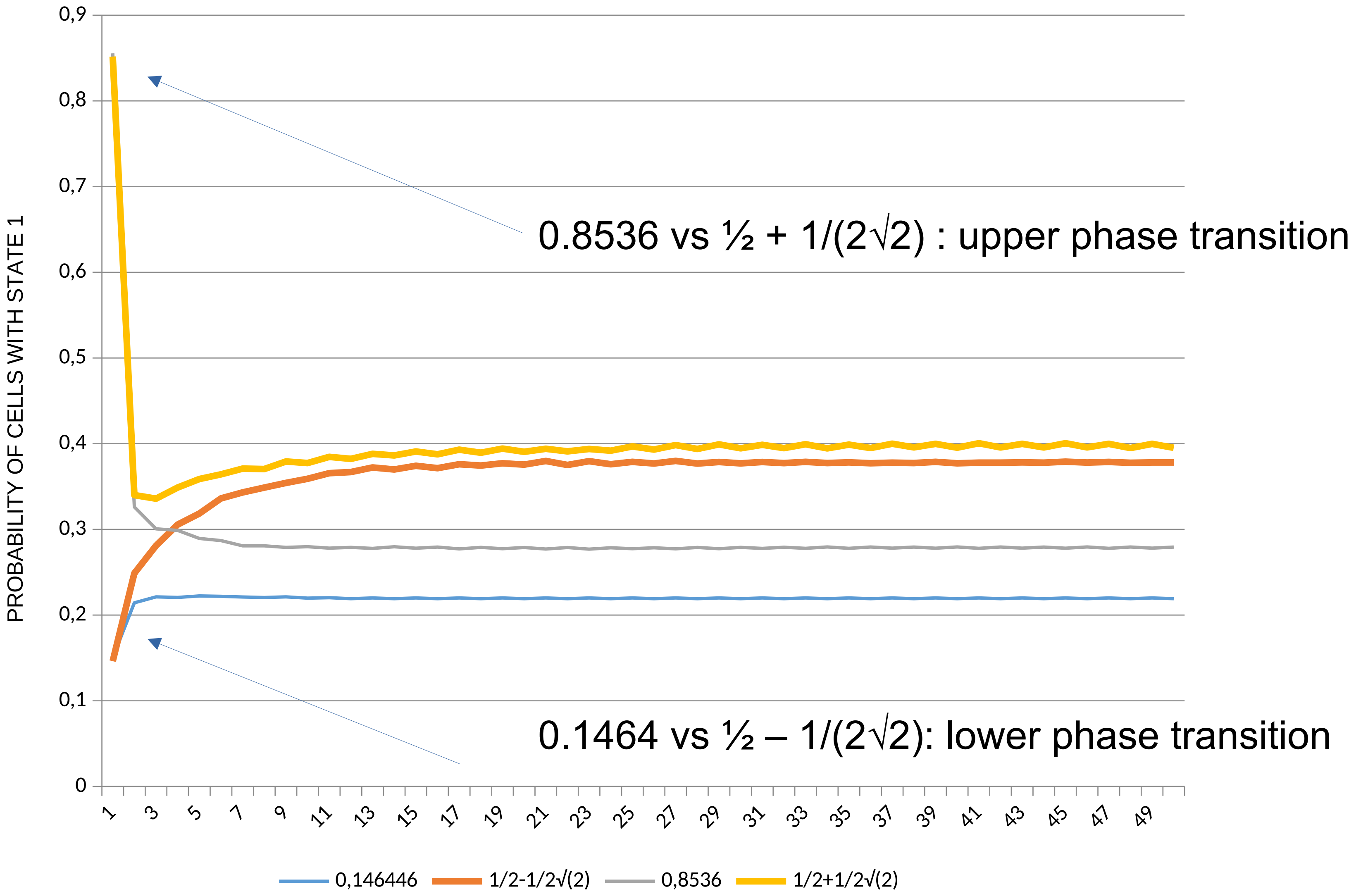
4. Uncritical state below ½ ± 1/(2√2)

5. Critical state with neighbor update, ½ – 1/(2√2)

6. Critical state with neighbor update, ½ + 1/(2√2)



First-Order Phase Transition



Trigonometric Expression of Criticality

$$\cos^2\left({}^{\pi}/_{8}\right) = \frac{1}{2} + \frac{1}{2\sqrt{2}}$$

$$\sin^2\left({}^{\pi}/_{8}\right) = \frac{1}{2} - \frac{1}{2\sqrt{2}}$$

$$\cos^2\left({}^{\pi}/_{8}\right) - \sin^2\left({}^{\pi}/_{8}\right) = \frac{1}{\sqrt{2}} = 2\sin\left({}^{\pi}/_{8}\right)\cos\left({}^{\pi}/_{8}\right)$$

General Expression:

$$a\cos^2 x - b\sin^2 x = \sin x \cos x$$

Trigonometric Expression of Criticality

divided by $\cos^2(x)$ on both sides:

$\therefore\ b\tan^2(x) + \tan(x) - a = 0$

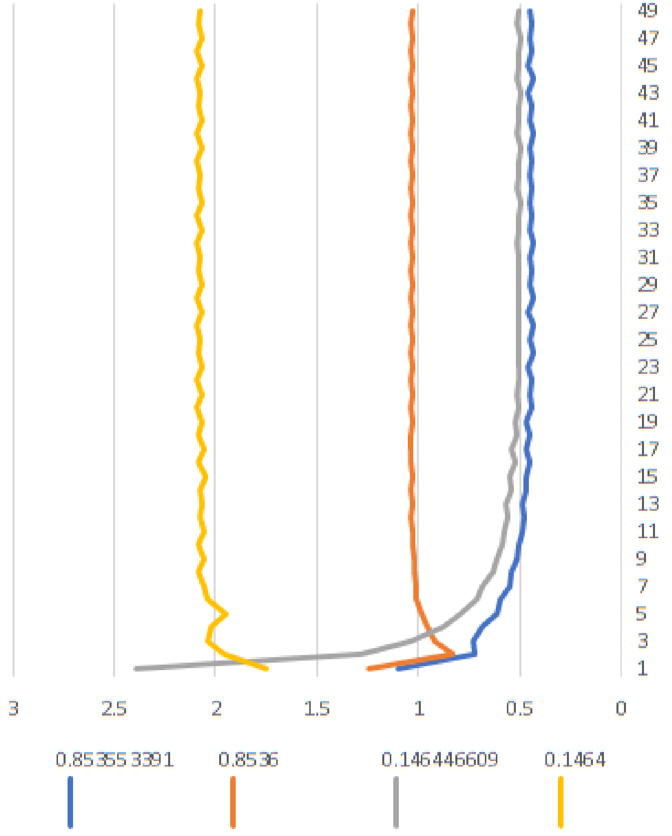
which corresponds to the code that draws the tangent equation.

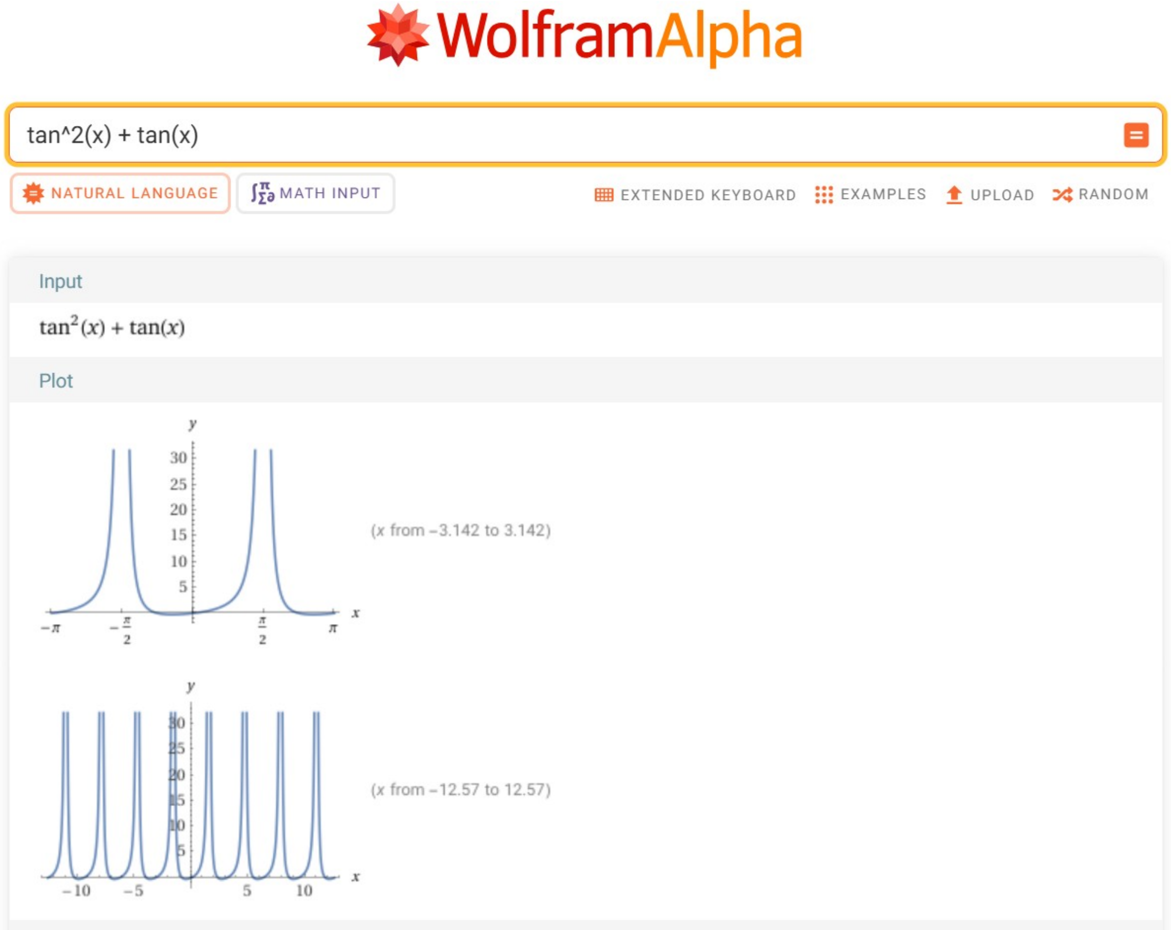
There are **no trigonometric functions** in the code!

Tangent Graph Output of CA

Critical and uncritical values are arms of the tangent graph.

$\tan^2 x$
$+ \tan x$
Graph





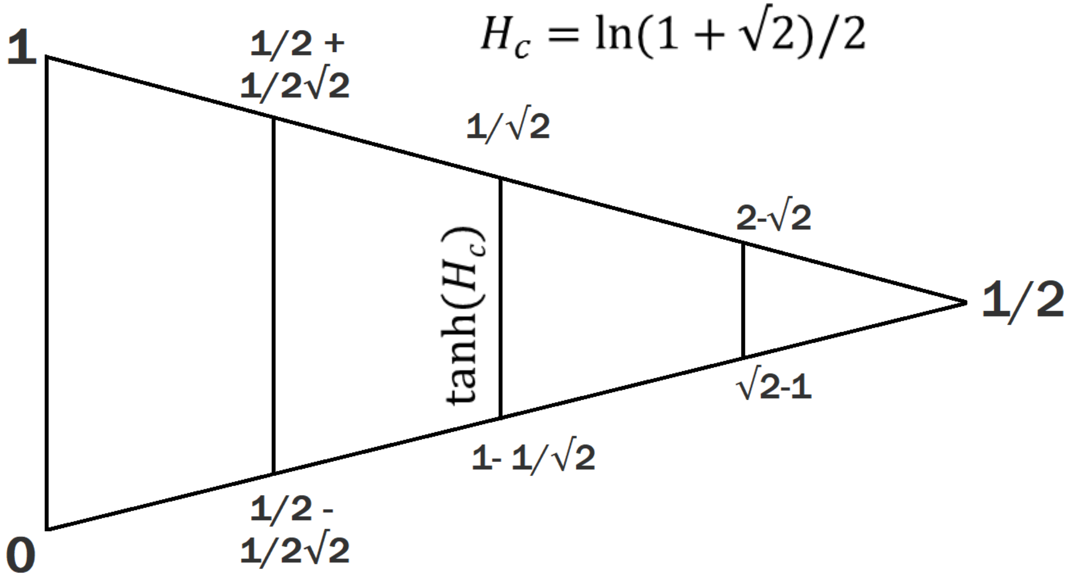
Ising Model Correspondence Visualized

Stereographic projection of 2D rectangular Ising model to 1D linear Ising chain

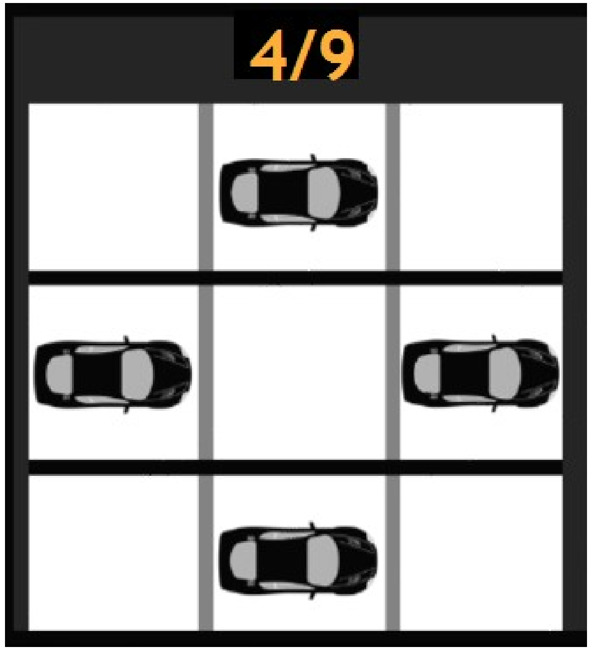
s = tan(1/2 * ϕ) = tan(1/2 * π/4)

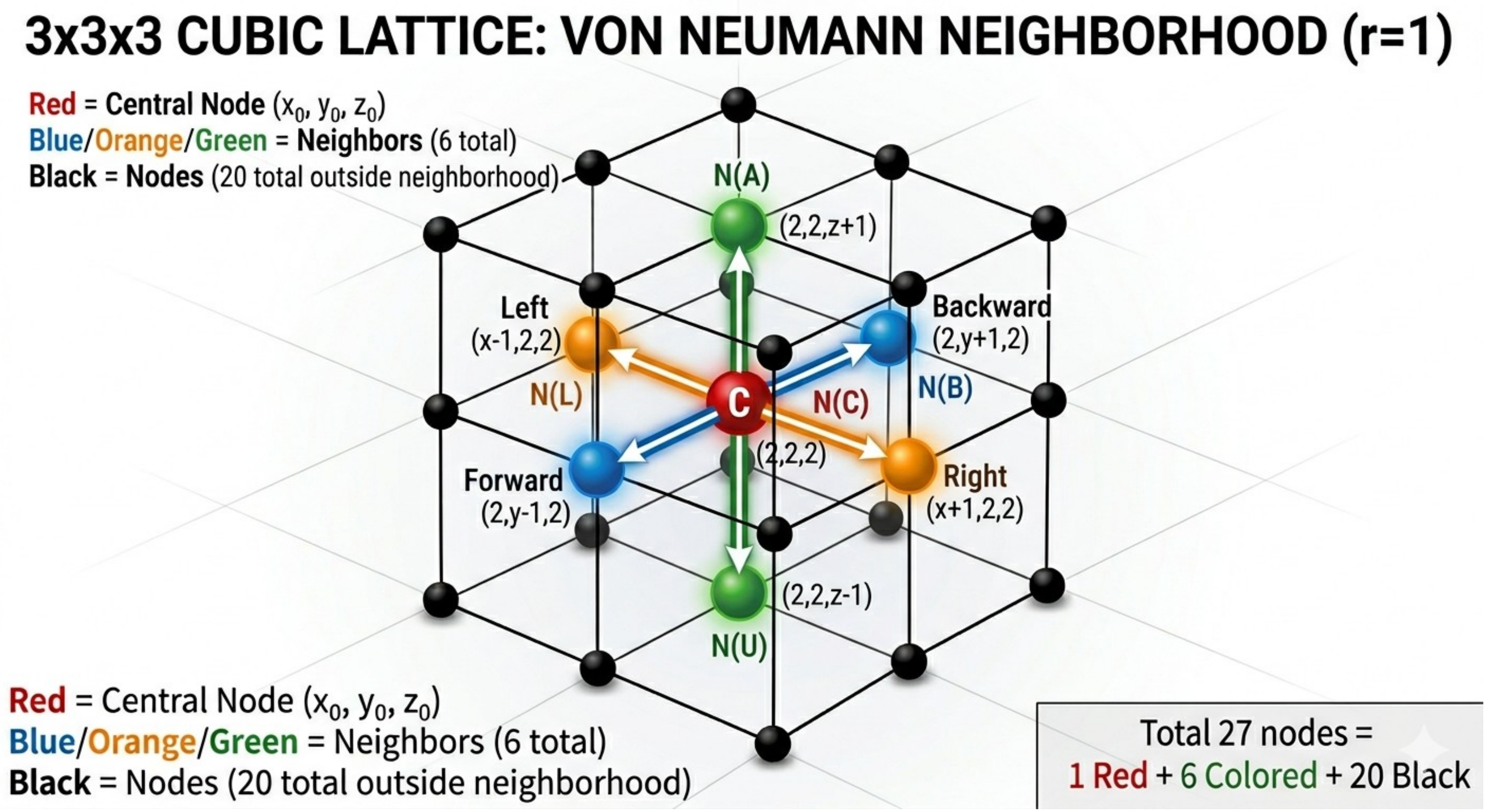
s = tanh(1/2 * ψ) = tanh(1/2 * ln(1+√2))

s = tan(π/8) = tanh(ln(1+√2)/2)



Von Neumann Neighborhoods 2D vs 3D





Approximate Inverse Ising critical temperature

d	Formula	Value	Known Ising Kc	Error
2	4/9	0.44444	0.44069	0.853%
3	6/27 = 4/18	0.22222	0.22165	0.258%
4	4/27	0.14815	0.14969	1.030%
5	4/36 = 1/9	0.11111	0.11395	2.491%
6	4/45	0.08889	0.09234	3.737%
7	4/54	0.07407	0.07768	4.642%
8	4/63	0.06349	0.06709	5.363%

$$K_d = 4 / (9 (d - 1))$$

Currently exploring into higher dimensions

github.com/goektug/Equation-Automata

github.com/goektug/Equation-Automata-3D